Answer Key - Chemistry - Undergraduate Level Assessment 23 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) Ti_3+; B) Cr_3+; C) Fe_3+; D) Zn_2+
   Note: Correct option: B - Cr_3+

2. **Answer:** C
   Options: A) 3; B) 2; C) 1; D) 0
   Note: Correct option: C - 1

3. **Answer:** C
   Options: A) K+; B) Ca_2+; C) Sc_3+; D) Rb+
   Note: Correct option: C - Sc_3+

4. **Answer:** C
   Options: A) Nuclear magnetic resonance; B) Infrared; C) Raman; D) Ultraviolet-visible
   Note: Correct option: C - Raman

5. **Answer:** C
   Options: A) all molecular bonds absorb IR radiation; B) IR peak intensities are related to molecular mass; C) most organic functional groups absorb in a characteristic region of the IR spectrum; D) each element absorbs at a characteristic wavelength
   Note: Correct option: C - most organic functional groups absorb in a characteristic region of the IR spectrum

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Neither position nor momentum can be known exactly.'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'MgO'.

Long Form Response

11. **Answer:** J = 1; u = 1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 345.0 MHz. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key